

Assignment on Future Value

Future Value

The amount to which the original amount will grow after earning interest is called future value.

$$\text{Future Value} = \text{Present Value} * (1 + r)^t$$

Questions:

Equal Cash Flows:

1. If you deposit \$10,000 in a bank account that pays 10% interest annually.
Required: How much will be in your account after 5 years?
2. Suppose that Peter Minuit did not become the first New York real estate tycoon, but instead had invested his \$24 at a 5 percent interest rate in New Amsterdam Savings Bank.
Required:
 - a. What would have been the balance in his account after 5 years?
 - b. After 50 years?
3. How much will a \$150,000 deposit in a bank yield after 13 years if the interest is paid at 6.5% annually?
Required: how much bank yield after 13 years?
4. Start-up Enterprises had sales last year of only \$500,000. However, a stock market analyst is bullish on the company and predicts that sales will double each year for 4 years.
Required: What are projected sales at the end of this period?
5. If \$25,400 is deposited in a bank, how much will the balance be after 7 years if the interest earned is 9.3% annually?
Required: How much will be balance after 7 years?

$$\begin{aligned}\text{Q1) future value} &= \text{Present value} \times (1+r)^t \\ &= 10,000 \left(1 + \frac{10}{100}\right)^5 \\ &= 16105.1\end{aligned}$$

$$\begin{aligned}\text{Q2) a) future value} &= \text{Present value} \times (1+r)^t \\ &= 24 \times \left(1 + \frac{5}{100}\right)^5 \\ &= 30.63\end{aligned}$$

b) after 50 years

$$\begin{aligned}\text{future value} &= \text{Present value} \times (1+r)^t \\ &= 24 \times \left(1 + \frac{5}{100}\right)^{50} \\ &= 275.21\end{aligned}$$

$$\begin{aligned}\text{Q3) future value} &= \text{Present value} \times (1+r)^t \\ &= 150,000 \left(1 + \frac{6.5}{100}\right)^{13} \\ &= 340123.12\end{aligned}$$

$$\begin{aligned}\text{Q4) Future value} &= \text{Present value} \times (1+r)^t \\ &= 500,000 \times \left(1 + \frac{200}{100}\right)^4 \\ &= 40500000\end{aligned}$$

$$\begin{aligned}\text{Q5) future value} &= \text{Present value} \times (1+r)^t \\ &= 25,400 \times \left(1 + \frac{9.3}{100}\right)^7 \\ &= 47334.17\end{aligned}$$

$$\begin{aligned}\text{Q6) future value} &= \text{Present value} \times (1+r)^t \\ &= 2500 \times \left(1 + \frac{8}{100}\right)^9 \\ &= 4447.51\end{aligned}$$

6. You deposit \$2,500 in your bank account. If the bank pays 8 percent simple interest.
Required:
 how much will you accumulate in your account after 9 years?

Multiple Cash Flows:

Activity 1: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 14%	Future Values
1	12,000	1.1400	13680
2	17,000	1.2996	22093.2
3	15,000	1.4815	22222.5
4	20,000	1.6890	33780

Activity 2: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 11%	Future Values
0	150,000	1	150,000
1	125,000	1.1100	138750
2	111,000	1.2321	136763.1
3	160,000	1.3676	218816
4	175,000	1.5181	265667.5
5	123,000	1.6861	207267.3
6	198,000	1.8704	370339.2

Activity 3: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 15%	Future Values
0	15,000	1	15 000
1	12,000	1.1500	13800
2	11,000	1.3225	14547.5
3	10,000	1.5209	15209
4	17,500	1.7490	30607.5
5	13,000	2.0114	26148.2
6	18,000	2.3131	41635.8
7	21,000	2.6600	55860
8	5,000	3.0590	15295

Activity 4: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 17%	Future Values
0	25,000	1	25000
1	35,000	1.17	40950
2	41,000	1.3689	56124.9
3	60,000	1.6016	96096
4	75,000	1.8738	140535
5	12,000	2.1924	26308.8

Activity 5: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 8%	Future Values
0	115,000	1	115000
1	127,000	1.0800	137160
2	116,000	1.1664	135302.4
3	130,000	1.2597	163761
4	175,000	1.3605	238087.5
5	130,000	1.4693	191009
6	181,000	1.5869	287228.9
7	213,000	1.7138	365039.4
8	125,000	1.8509	231362.5
9	165,000	1.9990	329835
10	200,000	2.1589	431780

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The amount to which the original amount will grow after earning interest is called future value.

$$\text{Future Value} = \text{Present Value} * (1 + r)^{t \rightarrow \text{time}}$$

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Questions:

Equal Cash Flows:

1. If you deposit \$10,000 in a bank account that pays 10% interest annually.

Required: How much will be in your account after 5 years?

$$\begin{aligned}\text{Future Value} &= 10,000 \times (1 + 10/100)^5 \\ &= 16105.1\end{aligned}$$

2. Suppose that Peter Minuit did not become the first New York real estate tycoon, but instead had invested his \$24 at a 5 percent interest rate in New Amsterdam Savings Bank.

Required:

- a. What would have been the balance in his account after 5 years?
b. After 50 years?

a) $\text{Future Value} = 24 \times (1 + 5/100)^5 = 30.630$

b) $\text{Future Value} = 24 \times (1 + 5/100)^{50} = 275.217$

3. How much will a \$150,000 deposit in a bank yield after 13 years if the interest is paid at 6.5% annually?

Required: how much bank yield after 13 years?

$$\begin{aligned}\text{Future Value} &= 150,000 \times (1 + \frac{6.5}{100})^{13} \\ &= 340123.1246\end{aligned}$$

4. Start-up Enterprises had sales last year of only \$500,000. However, a stock market analyst is bullish on the company and predicts that sales will double each year for 4 years.

Required: What are projected sales at the end of this period?

$$\begin{aligned}\text{Future Value} &= 500,000 \times 2^4 \quad (\text{cancel } 1 + \frac{2}{100}) \\ &= 500,000 \times 16 \\ &= 8,000,000\end{aligned}$$

5. If \$25,400 is deposited in a bank, how much will the balance be after 7 years if the interest earned is 9.3% annually?

Required: How much will be balance after 7 years?

$$\begin{aligned}\text{Future Value} &= 25,400 \times (1 + \frac{9.3}{100})^7 \\ &= 47460.64\end{aligned}$$

6. You deposit \$2,500 in your bank account. If the bank pays 8 percent simple interest.
Required:
 how much will you accumulate in your account after 9 years?

$$\text{Future Value} = 2500 \times \left(1 + \frac{8}{100}\right)^9$$

$$= 4997.511$$

Multiple Cash Flows:

Activity 1: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 14%	Future Values
1	12,000	1.1400	13680
2	17,000	1.2996	22093.2
3	15,000	1.4815	22222.5
4	20,000	1.6890	33780

Activity 2: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 11%	Future Values
0	150,000	1	150,000
1	125,000	1.1100	138750
2	111,000	1.2321	136753.1 136763.
3	160,000	1.3676	218816
4	175,000	1.5181	265667.5
5	123,000	1.6851	207267.3
6	198,000	1.8704	370339.2

Activity 3: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 15%	Future Values
0	15,000	1	15000
1	12,000	1.1500	13800
2	11,000	1.3225	14547.5
3	10,000	1.5209	15209
4	17,500	1.7490	30607.5
5	13,000	2.0114	26148.2
6	18,000	2.3131	41635.8
7	21,000	2.6600	55860
8	5,000	3.0590	15295

Activity 4: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 17%	Future Values
0	25,000	1	25000
1	35,000	1.17	40950
2	41,000	1.3689	56124.9
3	60,000	1.601613	96096.78
4	75,000	1.8738	140535
5	12,000	2.1924	26308.8

Activity 5: Calculate Future Values for the following Cash Flows

Years	Cash Flows	Future Value Factor @ 8%	Future Values
0	115,000	1	115000
1	127,000	1.0800	137160
2	116,000	1.1664	135302.4
3	130,000	1.2597	163761
4	175,000	1.3605	238087.5
5	130,000	1.4693	191009
6	181,000	1.5869	287228.9
7	213,000	1.7138	365039.4
8	125,000	1.8509	231362.5
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$$\text{Future Value} = \text{Present Value} * (1 + r)^t$$

Questions:

Equal Cash Flows:

1. If you deposit \$10,000 in a bank account that pays 10% interest annually.

Required: How much will be in your account after 5 years?

$$\begin{aligned} FV &= PV \times (1 + r)^t \\ &= 10000 \times (1 + 0.1)^5 \\ &= 16105.1 \end{aligned}$$

2. Suppose that Peter Minuit did not become the first New York real estate tycoon, but instead had invested his \$24 at a 5 percent interest rate in New Amsterdam Savings Bank.

Required:

- a. What would have been the balance in his account after 5 years?

- b. After 50 years?

$$\begin{aligned} a) FV &= P.V \times (1 + r)^t \\ &= 24 \times (1 + 0.05)^5 \\ &= 30.63 \end{aligned}$$

$$\begin{aligned} b) FV &= P.V \times (1 + r)^t \\ &= 24 \times (1 + 0.05)^{50} \\ &= 275.217 \end{aligned}$$

3. How much will a \$150,000 deposit in a bank yield after 13 years if the interest is paid at 6.5% annually?

Required: how much bank yield after 13 years?

$$\begin{aligned} FV &= PV \times (1 + r)^t \\ &= 150000 \times (1 + 0.065)^{13} \\ &= 340123.124 \end{aligned}$$

4. Start-up Enterprises had sales last year of only \$500,000. However, a stock market analyst is bullish on the company and predicts that sales will double each year for 4 years.

Required: What are projected sales at the end of this period?

$$\begin{aligned} FV &= PV \times (1 + r)^t \\ &= 500,000 \times (1 + 1)^4 \\ &= 8000000 \end{aligned}$$

5. If \$25,400 is deposited in a bank, how much will the balance be after 7 years if the interest earned is 9.3% annually?

Required: How much will be balance after 7 years?

$$\begin{aligned} FV &= PV \times (1 + r)^t \\ &= 25400 \times (1 + 0.093)^7 \\ &= 47334.179 \end{aligned}$$

6. You deposit \$2,500 in your bank account. If the bank pays 8 percent simple interest.

Required:

how much will you accumulate in your account after 9 years?

$$\begin{aligned}
 FV &= PV \times (1 + r)^t \\
 &= 2500 \times (1 + 0.08)^9 \\
 &= 4997.51
 \end{aligned}$$

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0	25,000	1	25000
1	35,000	1.2000	42000
2	41,000	1.4400	59040
3	60,000	1.7280	103680
4	75,000	2.0736	155520
5	12,000	2.4883	29859.6

Activity 5: Calculate Future Values for the following Cash Flows

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